

Small Counterexamples To Erdos Problem 97

Self-contained reviewer release note. Source-of-truth status remains README.md, STATE.md, RESULTS.md, docs/claims.md, and metadata/erdos97.yaml.

Status	Repo-local $n \leq 8$ finite-case release packet
Global status	Falsifiable/open; no general proof or counterexample claimed
Date	2026-07-09

Abstract

Erdos Problem #97 asks whether every convex polygon has a vertex with no other four vertices equidistant from it. This release note packages the repository's small-case theorem: no bad strictly convex polygon exists with $n \leq 8$. An elementary geometric proof is independently corroborated by the selected-witness machine-checked finite-case pipeline. The official problem remains falsifiable/open, and external review is still requested before paper-style citation.

Status And Scope

Official/global status: falsifiable/open, rechecked at <https://www.erdosproblems.com/97> on 2026-07-09; the page reports last edited 2025-10-27. This packet does not prove Erdos Problem #97, does not claim a counterexample, and does not promote any $n=9$ or $n=10$ artifact. It packages the $n \leq 8$ elementary proof and selected-witness finite-case evidence for external review.

Elementary Geometric Theorem

Theorem. If every vertex of a strictly convex n -gon has four other vertices at one common distance, then $n \geq 9$.

Count apex-marked isosceles triangles. Strict convexity permits at most one apex on either side of a base line, and no apex on the base itself; therefore $T \leq n(n-2)$. Badness gives $T \geq 6n$, excluding $n \leq 7$. In an octagon equality saturates every base capacity. Length-2 diagonals then force equal side lengths, while every length-3 diagonal forces one of two adjacent exterior turns to equal $2\pi/3$. These marked turns cover the 8-cycle, so at least four are required, contradicting total exterior turn 2π . The full proof is docs/n8-geometric-proof.md.

Problem And Selected Witnesses

A bad polygon would have vertices $p_0, \dots, p_{n-1}$ such that for every center i there are four other vertices at one common distance from p_i . Choosing one such 4-set S_i at every center gives the selected-witness formulation. Ruling out all selected-witness systems for a fixed n rules out bad polygons with that n , because any bad polygon supplies at least one selected row at each center.

Lemma Library L1-L10

- L1. Selected-witness extraction: every bad polygon supplies one selected 4-row S_i at each center.
- L2. Two-circle cap: for distinct centers a, b , $|S_a \cap S_b| \leq 2$, since two distinct circles share at most two points.
- L3. Witness-pair capacity: a fixed unordered witness pair can occur in at most two selected rows, because all such centers lie on the perpendicular bisector of that pair.
- L4. Crossing-bisector rule: if two rows share exactly two witnesses, the center chord is the perpendicular bisector of the common-witness chord, and the two chords cross at the midpoint.
- L5. Sharpened count: L2-L4 rule out selected-witness counterexamples for $n \leq 7$; the retained $n=7$ Fano enumeration independently audits the equality case.
- L6. $n=8$ indegree regularity: the witness-pair capacity forces every witness label to have selected indegree exactly 4.
- L7. $n=8$ incidence enumeration: with row 0 fixed to $\{1, 2, 3, 4\}$ by relabeling, exact enumeration leaves 15 canonical survivor classes.

L8. Forced geometry equations: two-overlap rows generate exact perpendicular-bisector equations; selected rows generate exact equal-distance equations.

L9. Survivor obstruction: class 12 has no compatible cyclic order; ten classes are killed by rational y_2 span certificates; classes 3, 4, 5, and 14 are killed by the named duplicate, collinearity, Groebner, and strict-interior certificates.

L10. Independent cross-checks: a SymPy-free rational recheck covers 11 classes, and an order-free z3 nonlinear-real-arithmetic replay finds all 15 classes UNSAT for strictly convex octagon realization.

Machine-Checked Corroboration

The selected-witness pipeline separately excludes $n \leq 8$. L5 handles $n \leq 7$. For $n=8$, L6 reduces the incidence layer to regular 4-in/4-out selected-witness systems. L7 exhausts the resulting necessary incidence systems and reduces them to 15 canonical classes. L8 translates every survivor into exact algebraic and cyclic-order constraints. L9 kills all 15 classes with exact checks. L10 gives two independent defensive replays. Therefore the repository has independent machine-checked finite-case corroboration for $n \leq 8$, subject to the stated artifact-review boundary.

Reproduction Commands

```
python scripts/independent_check_n8_artifacts.py --check --json
python scripts/enumerate_n8_incidence.py --summary
python scripts/analyze_n8_exact_survivors.py --check --json
python scripts/independent_n8_obstruction_recheck.py --check --json
python scripts/check_n8_class14_certificate.py --check --json
python scripts/check_n8_residual_certificates.py --check --json
python scripts/check_n8_survivors_smt.py --assert-clear --check-artifact
data/certificates/n8_survivors_smt.json
```

Expected Stable Invariants

The incidence enumeration reports 117072 balanced cap matrices with row 0 fixed, 4560 forced-perpendicular survivors with row 0 fixed, and 15 canonical survivor classes. The exact survivor checker rejects all 15 classes. The SymPy-free recheck kills classes 0,1,2,6,7,8,9,10,11,12,13 and explicitly leaves 3,4,5,14 to the focused Groebner-dependent audit path. The z3 SMT cross-check reports all 15 survivor classes UNSAT for strictly convex realization.

Reviewer Boundary

The reviewer should check the base-apex and diagonal-chain geometry, the row-0 relabeling symmetry break, the exact incidence filters, and the exact certificate replays for classes 3, 4, 5, and especially 14. A successful external review may support paper-style use of the $n \leq 8$ theorem and its computational corroboration. It would not update the official/global status and would not settle any case $n \geq 9$.